

Use Case Descriptions

Module: Authentication & Role Management

UC01 - Login / Logout

Use Case Name	Login / Logout	
Use Case ID	UC01	
Description	The system allows registered users to authenticate using their ID or email and password, and to securely log out, invalidating the active session.	
Priority	High	
Actor(s)	Admin, Office Staff, Programme Coordinator, Lecturer, Student, System	
Triggering Event	User wants to log into or log out of the system.	
Pre-condition	1. User has a registered account in the system. 2. Application is accessible online.	
Post-condition	1. User is successfully authenticated and a session is established (login). 2. Active session is invalidated and user is redirected to the login page (logout). 3. Login and logout activities are recorded in the audit log.	
Flow of Events	1.	User opens the login page.
	2.	System displays the login form with fields for ID/email and password.
	3.	User enters their credentials.
	4.	User submits the login form.
	5.	System validates the credentials against the user database.
	6.	System verifies that the account is active.
	7.	System generates a session token and loads the user's role and permissions.
	8.	System redirects the user to the role-based dashboard.
	9.	User performs system actions during the active session.

	10.	User clicks the Logout button.
	11.	System invalidates the session token and clears session data.
	12.	System redirects the user to the login page.
Alternative Flow	Alternative Flow - Forgot password	
	3a.	User clicks the Forgot Password link.
	3b.	System redirects to the password reset flow (UC02).
Exception Flow	Exception Flow 1 - Invalid credentials	
	5a.	System cannot match the credentials to any account.
	5b.	System displays an Invalid ID or password error.
	5c.	User retries with corrected credentials.
Exception Flow	Exception Flow 2 - Account disabled	
	6a.	System detects that the account is inactive or suspended.
	6b.	System displays a message indicating the account is disabled.
	6c.	System prevents the user from logging in.
Exception Flow	Exception Flow 3 - Session expired	
	9a.	User performs an action after session expiry.
	9b.	System displays a session expired message.
	9c.	System redirects the user to the login page.

UC02 - Reset Password

Use Case Name	Reset Password
Use Case ID	UC02
Description	The system allows users to reset their password through email-based recovery and enforces a password change on first-time login.
Priority	High
Actor(s)	Admin, Office Staff, Programme Coordinator, Lecturer, Student, System
Triggering Event	User has forgotten their password or is logging in for the first time.

Pre-condition	1. User has a registered account with a valid email address. 2. Application is accessible online.	
Post-condition	1. Password reset email is sent to the user's registered email. 2. User successfully sets a new password. 3. The new password is securely stored in the system.	
Flow of Events	1.	User clicks the Forgot Password link on the login page.
	2.	System displays the password reset request form.
	3.	User enters their registered email address.
	4.	User submits the request.
	5.	System validates the email and checks if it is registered.
	6.	System generates a unique password reset token with a time-limited expiry.
	7.	System sends the reset link to the user's registered email.
	8.	User opens the reset link and enters a new password.
	9.	System validates the new password against complexity requirements.
	10.	System updates the user's password securely.
	11.	System displays a success popup hint confirming the password reset.
Alternative Flow	Alternative Flow - First-time login forced password change	
	1a.	User logs in with the temporary password issued upon account creation.
	1b.	System detects that this is the first-time login.
	1c.	System redirects the user to the change-password page.
	1d.	User enters and confirms a new password.
	1e.	System validates and saves the new password.
	1f.	System logs the user into the dashboard.
Alternative Flow	Alternative Flow - Admin-assisted password reset	
	1a.	User contacts Admin to request a password reset.
	1b.	Admin opens the User Account Management module.
	1c.	Admin selects the user account and chooses the password reset option.

	1d.	System generates and issues a temporary password to the user.
	1e.	User logs in with the temporary password and is prompted to set a new password.
Exception Flow	Exception Flow 1 - Email not registered	
	5a.	System cannot find the email in the user database.
	5b.	System displays an Email not registered error message.
	5c.	User checks their email and tries again.
Exception Flow	Exception Flow 2 - Reset token expired	
	8a.	User attempts to use an expired reset link.
	8b.	System displays a Reset link expired message.
	8c.	User must request a new reset link.

UC03 - Manage User Account

Use Case Name	Manage User Account	
Use Case ID	UC03	
Description	The system allows the Admin to manage user accounts, including assigning one or more roles, configuring module permissions, activating, deactivating, or deleting user records.	
Priority	High	
Actor(s)	Admin, System	
Triggering Event	Admin wants to manage existing user accounts and permissions.	
Pre-condition	1. Admin is logged into the system. 2. Admin has permission to manage user accounts. 3. User accounts exist in the system.	
Post-condition	1. Selected user account is updated, activated, deactivated, or removed. 2. Updated role assignments and effective permissions are applied. 3. All changes are recorded in the audit log.	
Flow of Events	1.	Admin opens the User Account Management module.

	2.	System displays the list of existing user accounts with their roles and statuses.
	3.	Admin selects a user account to manage.
	4.	System displays the user account details with editable fields and action buttons.
	5.	Admin selects an action such as edit profile, change role, configure permissions, deactivate, or delete.
	6.	Admin enters or modifies the relevant information.
	7.	Admin confirms the changes.
	8.	System validates the changes.
	9.	System applies the changes to the user account.
	10.	System displays a success popup hint.
Alternative Flow	Alternative Flow - Assign multiple roles	
	5a.	Admin selects the option to assign multiple roles to the user.
	5b.	System displays the list of available roles with checkboxes.
	5c.	Admin selects one or more roles (e.g., Lecturer and Programme Coordinator).
	5d.	System computes the union of permissions across selected roles.
	5e.	System saves the new role assignments.
Exception Flow	Exception Flow 1 - Cannot delete own account	
	5a.	Admin attempts to delete their own account.
	5b.	System displays a Cannot delete your own account error message.
	5c.	System prevents the action.
Exception Flow	Exception Flow 2 - Conflicting permissions	
	8a.	System detects a conflicting permission rule.
	8b.	System displays a warning with conflict details.
	8c.	Admin reviews and adjusts the permission settings.

UC04 - Register User Account

Use Case Name	Register User Account	
Use Case ID	UC04	
Description	The system allows Admin or Office Staff to register new user accounts individually for late registrations, automatically generating temporary passwords and emailing credentials to the new user.	
Priority	High	
Actor(s)	Admin, Office Staff, System	
Triggering Event	Admin or Office Staff wants to create a new user account.	
Pre-condition	1. User is logged into the system. 2. User has permission to register new accounts. 3. New user's information (name, ID, email, role) is available.	
Post-condition	1. A new user account is created with a system-generated temporary password. 2. Login credentials are emailed to the new user's registered email address. 3. Account creation is recorded in the audit log.	
Flow of Events	1.	User opens the User Account Management module.
	2.	User selects the option to register a new account.
	3.	System displays the registration form.
	4.	User enters the new user's name, ID, email, and selects the role(s).
	5.	User confirms the registration.
	6.	System validates the entered information.
	7.	System auto-generates a secure temporary password.
	8.	System creates the new user record and assigns role-based permissions.
	9.	System sends an email containing the login credentials to the new user.
	10.	System displays a success popup hint confirming the registration.
Alternative Flow	Alternative Flow - Late registration with multi-role assignment	

	4a.	User assigns multiple roles to the new account during creation.
	4b.	System applies the union of permissions for the assigned roles.
Exception Flow	Exception Flow 1 - Duplicate user account	
	6a.	System detects an existing account with the same ID or email.
	6b.	System displays an error message indicating the duplicate.
	6c.	System prevents the new account from being created.
	6d.	User reviews the existing account or modifies the input.
Exception Flow	Exception Flow 2 - Email delivery failure	
	9a.	System fails to send the credentials email.
	9b.	System notifies the registering user of the failure.
	9c.	User can choose to resend the email or copy the credentials manually.

Module: Student Registry

UC05 - Bulk Import Students

Use Case Name	Bulk Import Students
Use Case ID	UC05
Description	The system allows Admin or Office Staff to bulk import student data from Excel or CSV files, automatically creating student accounts with temporary passwords during the import.
Priority	High
Actor(s)	Admin, Office Staff, System
Triggering Event	User wants to import multiple student records and create their accounts in a single operation.
Pre-condition	1. User is logged into the system. 2. User has permission to import student data.

	3. Source Excel or CSV file is prepared in the required template format.	
Post-condition	1. Valid student records are added to the Student Registry. 2. Student accounts are created with temporary passwords. 3. Login credentials are emailed to each newly imported student. 4. Import action is recorded in the audit log.	
Flow of Events	1.	User opens the Student Registry module.
	2.	User clicks the Bulk Import button.
	3.	System displays the file upload interface with a template download link.
	4.	User uploads the Excel or CSV file.
	5.	System validates the file format and column headers.
	6.	System parses the file row by row.
	7.	System validates each row against required fields and unique constraints.
	8.	System creates student profile records and auto-generates accounts with temporary passwords.
	9.	System emails credentials to each newly created student.
	10.	System generates an import summary report.
	11.	System displays a success popup hint with the number of successful and failed imports.
Alternative Flow	Alternative Flow - Download import template	
	3a.	User clicks the template download link.
	3b.	System provides a downloadable Excel or CSV template.
	3c.	User fills the template with student data and uploads it.
Exception Flow	Exception Flow 1 - Invalid file format	
	5a.	System detects an unsupported file format.
	5b.	System displays an Only XLSX or CSV files allowed error.
	5c.	User must upload a valid file format.
Exception Flow	Exception Flow 2 - Duplicate student records	
	7a.	System detects rows with student IDs that already exist.
	7b.	System skips the duplicate records and logs them.

	7c.	System continues importing the remaining valid records.
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UC06 - Maintain Student Profile

Use Case Name	Maintain Student Profile	
Use Case ID	UC06	
Description	The system allows authorized users to maintain master student profile records and link each profile to related supervisor appointments, panel appointments, marks, files, letters, and announcements.	
Priority	High	
Actor(s)	Office Staff, System	
Triggering Event	Office Staff wants to create, view, or update a student profile record.	
Pre-condition	1. Office Staff is logged into the system. 2. Office Staff has permission to manage student profiles.	
Post-condition	1. Student profile record is created or updated in the Student Registry. 2. Profile is automatically linked to its related records across modules. 3. Profile changes are recorded in the audit log.	
Flow of Events	1.	Office Staff opens the Student Registry module.
	2.	System displays the list of existing student records.
	3.	Office Staff selects a student record to update or chooses to create a new record.
	4.	System displays the student profile form with editable fields.
	5.	Office Staff enters or modifies student information such as name, ID, programme, intake batch, and status.
	6.	Office Staff confirms the changes.
	7.	System validates the entered information.
	8.	System saves the student profile record and updates linked records across modules.
	9.	System displays a success popup hint.

Alternative Flow	Alternative Flow - Update student academic status	
	5a.	Office Staff selects the option to update the student's academic status.
	5b.	System displays the status update form with current status and dropdown options.
	5c.	Office Staff selects the new status and adds an optional note.
	5d.	System validates the status transition.
	5e.	System updates the student's status and logs the change.
Exception Flow	Exception Flow 1 - Duplicate student ID	
	7a.	System detects an existing student record with the same ID.
	7b.	System displays a Duplicate student ID error message.
	7c.	Office Staff reviews the existing record or modifies the input.

UC07 - Search and Filter Students

Use Case Name	Search and Filter Students	
Use Case ID	UC07	
Description	The system allows authorized users to search and filter student records by programme, semester, supervisor, or other available criteria.	
Priority	Medium	
Actor(s)	Office Staff, Programme Coordinator, Lecturer, System	
Triggering Event	User wants to find specific student records using search or filter criteria.	
Pre-condition	1. User is logged into the system. 2. User has permission to view student records. 3. Student records exist in the Student Registry.	
Post-condition	1. List of matching student records is displayed. 2. User can select a record to view its full details.	
Flow of Events	1.	User opens the Student Registry module.

	2.	System displays the search and filter interface.
	3.	User enters a search keyword or selects filter criteria such as programme, semester, or supervisor.
	4.	User submits the search or filter request.
	5.	System queries the Student Registry with the specified criteria.
	6.	System retrieves matching student records.
	7.	System displays the filtered student list.
	8.	User reviews the results or selects a record for details.
Alternative Flow	Alternative Flow - Reset search and filters	
	4a.	User clicks the Reset button to clear all search and filter inputs.
	4b.	System clears all criteria and displays the full student list.
Exception Flow	Exception Flow 1 - No matching records found	
	6a.	System finds no records matching the search or filter criteria.
	6b.	System displays a No matching students found message.
	6c.	User adjusts the criteria and tries again.

UC08 - View and Update Profile

Use Case Name	View and Update Profile
Use Case ID	UC08
Description	The system allows students to view their own profile information and records linked to their profile, and to update permitted contact details such as email and phone number.
Priority	High
Actor(s)	Student, System
Triggering Event	Student wants to check or update their personal profile information.
Pre-condition	1. Student is logged into the system. 2. Student profile exists in the Student Registry.

Post-condition	1. Student profile information is displayed. 2. Permitted contact details are updated when changes are submitted. 3. Profile update is recorded in the audit log.	
Flow of Events	1.	Student opens the Profile page.
	2.	System retrieves the student profile linked to the logged-in account.
	3.	System displays profile information and linked records.
	4.	Student selects Edit for permitted fields.
	5.	Student updates allowed contact details such as email or phone.
	6.	System validates the updated information.
	7.	System saves the update.
	8.	System displays a success popup hint.
Alternative Flow	Alternative Flow - View only	
	4a.	Student only views the profile and makes no changes.
	4b.	System leaves the profile unchanged.
Exception Flow	Exception Flow 1 - Invalid contact format	
	6a.	System detects invalid email or phone number format.
	6b.	System displays the validation issue.
	6c.	Student corrects the information and submits again.

UC09 - View Linked Records

Use Case Name	View Linked Records
Use Case ID	UC09
Description	The system allows students to view records linked to their own profile, including supervisor appointments, panel appointments, marks, files, letters, and announcements.
Priority	High
Actor(s)	Student, System
Triggering Event	Student wants to review records related to their own profile.

Pre-condition	1. Student is logged into the system. 2. Student profile exists. 3. Linked records exist or can be queried by the system.	
Post-condition	1. Linked records belonging to the student are displayed. 2. Student can open allowed record details.	
Flow of Events	1.	Student opens the Linked Records page.
	2.	System identifies the logged-in student's profile.
	3.	System displays linked record categories.
	4.	Student selects a category.
	5.	System retrieves records under the selected category.
	6.	System displays the records in a structured table.
	7.	Student selects a record to view more details.
Alternative Flow	Alternative Flow - View all linked records summary	
	3a.	Student selects the option to view a summary of all linked record categories.
	3b.	System displays a consolidated summary.
	3c.	Student selects a category to drill down into details.
Exception Flow	Exception Flow 1 - No linked records	
	5a.	System finds no records of the selected category for the student.
	5b.	System displays a No records found message.
	5c.	Student selects a different category or returns to the profile.

Module: Supervisor Appointment

UC10 - Submit Supervisor Appointment Request

Use Case Name	Submit Supervisor Appointment Request
Use Case ID	UC10

Description	The system allows students to submit supervisor appointment requests with proposed supervisor and research details for review by the postgraduate office.	
Priority	High	
Actor(s)	Student, System	
Triggering Event	Student wants to propose a supervisor for their research project.	
Pre-condition	1. Student is logged into the system. 2. Student profile exists. 3. Supervisor request submission period is open. 4. Lecturer records are available.	
Post-condition	1. Supervisor appointment request is created. 2. Request status is set to Pending. 3. Request is queued for processing by Office Staff.	
Flow of Events	1.	Student opens the Supervisor Appointment Request page.
	2.	System displays the request form.
	3.	Student enters proposed supervisor details, research title, semester, and supporting notes.
	4.	Student attaches supporting documents if required.
	5.	System validates required fields and uploaded files.
	6.	Student confirms submission.
	7.	System creates the supervisor appointment request with Pending status.
	8.	System displays a success popup hint and shows the request status.
Alternative Flow	Alternative Flow - Save as draft	
	5a.	Student saves the request as a draft instead of submitting.
	5b.	System stores the draft without routing it for processing.
	5c.	Student can return later to complete and submit the draft.
Exception Flow	Exception Flow 1 - Missing required fields	
	5a.	System detects missing required fields or invalid lecturer details.
	5b.	System highlights the missing fields and displays an error.

	5c.	Student completes the required fields and resubmits.
Exception Flow	Exception Flow 2 - Duplicate active request	
	5a.	System detects an existing active request from the same student.
	5b.	System displays a Duplicate request error message.
	5c.	System prevents submission.

UC11 - Track Supervisor Appointment Request

Use Case Name	Track Supervisor Appointment Request	
Use Case ID	UC11	
Description	The system allows students to track the approval status of their submitted supervisor appointment requests.	
Priority	High	
Actor(s)	Student, System	
Triggering Event	Student wants to check the status of a submitted supervisor appointment request.	
Pre-condition	1. Student is logged into the system. 2. Student has at least one supervisor appointment request.	
Post-condition	1. Request status and any approver comments are displayed.	
Flow of Events	1.	Student opens the Supervisor Request Status page.
	2.	System retrieves the student's supervisor appointment requests.
	3.	System displays request status, approver comments, and the last updated date.
	4.	Student selects a request to view details.
	5.	System displays the detailed status history and any comments from approvers.
Alternative Flow	Alternative Flow - View rejected request reason	
	4a.	Student selects a rejected request.
	4b.	System displays the rejection reason and approver comment.

	4c.	Student reviews the reason and may submit a new request.
Exception Flow	Exception Flow - No request found	
	2a.	System finds no supervisor appointment requests for the student.
	2b.	System displays a No requests found message.

UC12 - Process Supervisor Appointment Request

Use Case Name	Process Supervisor Appointment Request	
Use Case ID	UC12	
Description	The system automatically processes a submitted supervisor appointment request by validating required fields, recording proposed supervisor details, research title, semester, and supporting notes.	
Priority	High	
Actor(s)	System	
Triggering Event	A student submits a supervisor appointment request.	
Pre-condition	1. Supervisor appointment request has been submitted. 2. Student and lecturer records are available. 3. Approval workflow is configured.	
Post-condition	1. Supervisor appointment request is processed. 2. Proposed supervisor details, research title, semester, and supporting notes are recorded. 3. Duplicate request checking is triggered. 4. Supervisor selection validation is triggered. 5. Request is ready for approval routing if all validation checks pass.	
Flow of Events	1.	System receives the submitted supervisor appointment request.
	2.	System retrieves the submitted student information, proposed supervisor details, research title, semester, and supporting notes.
	3.	System checks that all required submission fields are completed.

	4.	System validates the proposed supervisor against lecturer records.
	5.	System creates the supervisor appointment record with initial status.
	6.	System triggers duplicate supervisor request checking.
	7.	System triggers supervisor selection validation.
	8.	System updates the request status based on the validation result.
	9.	System displays the request processing result.
Alternative Flow	None	
Exception Flow	Exception Flow - Missing required details	
	3a.	System detects missing required fields.
	3b.	System displays an error message showing the missing fields.
	3c.	System prevents the supervisor appointment request from being processed.

UC13 - View Supervisor Appointment Records

Use Case Name	View Supervisor Appointment Records
Use Case ID	UC13
Description	The system allows authorized users to view supervisor appointment records filtered by status, student, lecturer, or semester.
Priority	High
Actor(s)	Office Staff, Programme Coordinator, Lecturer (Supervisor), Student, System
Triggering Event	User wants to view supervisor appointment records.
Pre-condition	1. User is logged into the system. 2. Supervisor appointment records exist. 3. User has permission to view supervisor appointment records.

Post-condition	1. Supervisor appointment records matching the filter are displayed.	
Flow of Events	1.	User opens the Supervisor Appointment Records page.
	2.	System identifies the user's role and access permission.
	3.	System displays the filter interface based on the user's access scope.
	4.	User applies filters such as status, student, lecturer, semester, or programme.
	5.	System retrieves matching supervisor appointment records based on the applied filters and the user's role-based access permission.
	6.	System displays the accessible records in a structured list.
	7.	User selects a record to view its details.
	8.	System displays the selected record details if the user is authorized to access it.
Alternative Flow	Alternative Flow - Search by keyword	
	4a.	User enters a keyword in the search box.
	4b.	System searches records matching the keyword across student, lecturer, and research title, limited by the user's role-based access permission.
	4c.	System displays the matching records.
Exception Flow	Exception Flow 1 - No matching records	
	5a.	System finds no matching records.
	5b.	System displays a No matching records message.
	Exception Flow 2 - Unauthorized record access	
	8a.	User attempts to access a supervisor appointment record outside their permission scope.
	8b.	System denies access to the record.
	8c.	System displays an unauthorized access message.

UC14 - Detect Duplicate Supervisor Requests

Use Case Name	Detect Duplicate Supervisor Requests
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Use Case ID	UC14	
Description	The system detects duplicate or conflicting supervisor requests during submission or processing.	
Priority	High	
Actor(s)	Student, System	
Triggering Event	A supervisor appointment request is submitted or being processed.	
Pre-condition	1. Student is logged into the system. 2. Student has filled in the supervisor appointment request form. 3. Existing supervisor appointment records are available for comparison.	
Post-condition	1. Duplicate or conflicting requests are identified. 2. The request is either allowed to proceed or rejected.	
Flow of Events	1.	Student fills in the supervisor appointment request form.
	2.	Student clicks the submit button.
	3.	System retrieves the student ID, proposed supervisor, and semester from the request.
	4.	System compares the request with existing supervisor appointment records.
	5.	System checks whether the same student has already submitted a supervisor request for the same semester.
	6.	System checks whether the request conflicts with an existing approved or pending supervisor appointment.
	7.	System displays the duplicate or conflict checking result.
	8.	System allows the request to be submitted if no duplicate or conflict is found.
Alternative Flow	Alternative Flow - Duplicate or conflict found	
	6a.	System detects a possible duplicate or conflict.
	6b.	System displays the conflicting existing record.
	6c.	System displays the relevant existing request or appointment record.

	6d.	System prevents duplicate submission
Exception Flow	Exception Flow - Existing records cannot be retrieved	
	3a.	System fails to retrieve existing records for comparison.
	3b.	System displays an error message and aborts the duplicate check.

UC15 - Validate Supervisor Selection

Use Case Name	Validate Supervisor Selection	
Use Case ID	UC15	
Description	The system validates the proposed supervisor selection against existing lecturer records and current workload to ensure validity and feasibility.	
Priority	High	
Actor(s)	System, Student	
Triggering Event	Supervisor appointment request is submitted or being processed.	
Pre-condition	1. Supervisor appointment request exists. 2. Lecturer records and workload data are accessible.	
Post-condition	1. Supervisor selection is validated as feasible or flagged for review.	
Flow of Events	1.	System receives the supervisor appointment request for validation.
	2.	System retrieves the proposed supervisor's lecturer record.
	3.	System checks the lecturer's current supervision workload for the semester.
	4.	System verifies the lecturer's availability and eligibility.
	5.	System displays the validation result.
Alternative Flow	Alternative Flow - Workload at capacity	
	3a.	System detects that the proposed supervisor is at capacity.
	3b.	System flags the request and displays a warning.

	3c.	System allows the request to be submitted only after the student confirms the warning by clicking “Yes”.
Exception Flow	Exception Flow - Invalid supervisor selection	
	2a.	System cannot find the proposed lecturer in the records.
	2b.	System displays an Invalid supervisor error.

UC16 - Track Supervisor Workload

Use Case Name	Track Supervisor Workload	
Use Case ID	UC16	
Description	The system tracks the number of times each lecturer is selected as supervisor per semester to support workload monitoring and balancing.	
Priority	Medium	
Actor(s)	Student, Lecturer, Programme Coordinator, Office Staff, System	
Triggering Event	User wants to review supervisor workload for the current or past semesters.	
Pre-condition	1. User is logged into the system. 2. Supervisor appointment records exist.	
Post-condition	1. Supervisor workload data is displayed by lecturer and semester 2. Students can use the workload count to support supervisor selection. .	
Flow of Events	1.	User opens the Supervisor Workload page.
	2.	System identifies the user’s role and access permission.
	3.	System displays the workload dashboard with lecturers and their assigned student counts.
	4.	User filters by semester, department, or lecturer.
	5.	System retrieves matching workload data based on the user’s access permission.
	6.	System displays the workload distribution.
	7.	User reviews the data.
Alternative Flow	None	

Exception Flow	Exception Flow - No workload data	
	4a.	System finds no data for the selected semester.
	4b.	System displays a No workload data message.

UC17 - Generate Supervisor Confirmation Letter

Use Case Name	Generate Supervisor Confirmation Letter	
Use Case ID	UC17	
Description	The system auto-generates a supervisor appointment confirmation letter upon approval and delivers it to the student and the assigned supervisor within the system.	
Priority	High	
Actor(s)	System, Student, Lecturer (Supervisor)	
Triggering Event	A supervisor appointment request is approved through the workflow.	
Pre-condition	1. Supervisor appointment request has been approved. 2. Student and supervisor records exist. 3. Supervisor confirmation letter template is available.	
Post-condition	1. Supervisor confirmation letter is generated and saved to the document repository. 2. Letter is delivered to the student and supervisor within the system. 3. Both recipients receive a notification.	
Flow of Events	1.	System detects that the supervisor appointment request has been approved.
	2.	System retrieves the supervisor confirmation letter template.
	3.	System auto-fills the template with student and supervisor details.
	4.	System generates the letter as a PDF file.
	5.	System saves the letter to the document repository.
	6.	System links the letter to the student and supervisor records.
	7.	System delivers the letter notification to both recipients.

	8.	Recipients can preview or download the letter from their dashboard.
Alternative Flow	Alternative Flow - Manual regeneration by Office Staff	
	1a.	Office Staff manually triggers letter regeneration after data correction.
	1b.	System regenerates the letter with updated data.
	1c.	System replaces the previous version and notifies recipients.
Exception Flow	Exception Flow - Template missing or invalid	
	2a.	System cannot find the supervisor confirmation letter template.
	2b.	System logs the error and notifies Admin.
	2c.	System aborts the letter generation.

Module: Panel Appointment

UC18 - Recommend Panel Member

Use Case Name	Recommend Panel Member	
Use Case ID	UC18	
Description	The system allows supervisors to recommend panel members for coursework projects.	
Priority	High	
Actor(s)	Lecturer (Supervisor), System	
Triggering Event	Supervisor wants to propose panel members for their student's coursework project.	
Pre-condition	1. Supervisor is logged into the system. 2. Student has an approved supervisor appointment. 3. Lecturer records are available.	
Post-condition	1. Panel recommendation is created and queued for coordinator review.	
Flow of Events	1.	Supervisor opens the Panel Recommendation page.

	2.	Supervisor selects the student.
	3.	System displays the panel recommendation form.
	4.	Supervisor selects proposed panel members from the lecturer list.
	5.	Supervisor adds justification notes.
	6.	System validates the selection.
	7.	Supervisor submits the recommendation.
	8.	System routes the recommendation according to the approval workflow
	9.	System displays a success popup hint.
Alternative Flow	Alternative Flow - Save as draft	
	6a.	Supervisor saves the recommendation as draft.
	6b.	System stores the draft for later submission.
Exception Flow	Exception Flow - Invalid panel selection	
	6a.	System detects an invalid panel member (e.g., the supervisor themselves).
	6b.	System displays an invalid panel selection error.
	6c.	Supervisor corrects the selection.

UC19 - Validate Panel Appointment

Use Case Name	Validate Panel Appointment
Use Case ID	UC19
Description	The system validates panel appointment rules during panel recommendation and prevents duplicate or invalid panel assignments before the recommendation is submitted.
Priority	High
Actor(s)	Lecturer (Supervisor), System
Triggering Event	Supervisor selects or submits proposed panel members for a student's coursework project.
Pre-condition	1. Panel recommendation exists. 2. Existing panel appointment records are accessible.

Post-condition	1. Panel appointment recommendation is validated. 2. Duplicate or invalid panel assignments are flagged. 3. Valid recommendation can proceed for submission and approval routing.	
Flow of Events	1.	Supervisor selects proposed panel members in the Panel Recommendation page.
	2.	System checks each proposed panel member against existing panel appointment records.
	3.	System verifies that the proposed panel member is not the supervisor of the same student.
	4.	System checks panel appointment rules, such as duplicate assignment, lecturer eligibility, and workload limit.
	5.	System displays the validation result to the Supervisor.
	6.	Supervisor reviews the validation result.
	7.	If no issue is found, Supervisor submits the panel recommendation.
	8.	System allows the recommendation to proceed to the approval workflow.
Alternative Flow	None.	
Exception Flow	Exception Flow - Rule violation detected	
	4a.	System detects a rule violation.
	4b.	System flags the recommendation and displays a warning.
	4c.	Coordinator decides whether to override or reject.

UC20 - Search Panel Appointment History

Use Case Name	Search Panel Appointment History
Use Case ID	UC20
Description	The system allows users to search past panel appointment records and approval history.
Priority	Medium
Actor(s)	Programme Coordinator, Office Staff, Lecturer(Panel), System
Triggering Event	User wants to look up past panel appointment records.

Pre-condition	1. User is logged into the system. 2. Panel appointment records exist. 3. User has permission to search panel appointment records.	
Post-condition	1. Matching panel appointment records are displayed according to the user's role-based access permission.	
Flow of Events	1.	User opens the Panel Appointment History page.
	2.	System identifies the user's role and access permission.
	3.	System displays the search and filter interface.
	4.	User enters search criteria such as student, lecturer, semester, programme, or status.
	5.	User submits the search.
	6.	System retrieves matching panel appointment records based on the search criteria and the user's role-based access permission.
	7.	System displays the accessible records in a structured list.
	8.	User selects a record to view its full details.
	9.	System displays the panel appointment details if the user is authorized to access the selected record.
Alternative Flow	Alternative Flow - Export search results	
	7a.	User clicks the Export button to export the results.
	7b.	System exports only the panel appointment records that the user is authorized to view.
	7c.	System provides the results as a downloadable file.
Exception Flow	Exception Flow 1 - No matching records	
	6a.	System finds no matching records.
	6b.	System displays a No matching records message.
	Exception Flow 2 - Unauthorized record access	
	9a.	User attempts to access a panel appointment record outside their permission scope.
	9b.	System denies access to the record.
	9c.	System displays an unauthorized access message.

UC21 - Track Panel Member Selection Count

Use Case Name	Track Panel Member Selection Count	
Use Case ID	UC21	
Description	The system tracks the number of times each lecturer is selected as panel member per semester to support workload monitoring.	
Priority	Medium	
Actor(s)	Office Staff, Programme Coordinator, Lecturer, System	
Triggering Event	User wants to review panel member selection counts.	
Pre-condition	1. User is logged into the system. 2. User is an authorized staff user. 3. User is not a student. 4. Panel appointment records exist.	
Post-condition	1. Panel member selection counts are displayed by lecturer and semester. 2. User can review panel workload distribution before making or reviewing panel appointment decisions.	
Flow of Events	1.	User opens the Panel Member Selection Count page.
	2.	System verifies that the user is an authorized non-student user.
	3.	System displays the count dashboard with lecturers and their panel appointment counts.
	4.	User filters the records by semester, department, or lecturer.
	5..	System retrieves matching count data.
	6.	System displays the panel workload distribution.
	7.	User reviews the data.
Alternative Flow	None	
Exception Flow	Exception Flow 1 - No data for semester	
	5a.	System finds no panel data for the selected semester.
	5b.	System displays a No data message.
	Exception Flow 2 - Unauthorized student access	
	2a.	System detects that the user is a student.

	2b.	System denies access to the panel member selection count page.
	2c.	System displays an unauthorized access message.

UC22 - View Panel Information

Use Case Name	View Panel Information	
Use Case ID	UC22	
Description	The system allows students to view their approved panel members and evaluation task information related to their own coursework project.	
Priority	Medium	
Actor(s)	Student, System	
Triggering Event	Student wants to know the panel members and evaluation arrangement for their project.	
Pre-condition	1. Student is logged into the system. 2. Panel appointment record exists for the student. 3. Record has been approved.	
Post-condition	1. Approved panel and evaluation information is displayed to the student.	
Flow of Events	1.	Student opens the Panel Information page.
	2.	System identifies the student's project record.
	3.	System retrieves approved panel appointment and evaluation task information.
	4.	System displays panel member names, roles, evaluation stage, and semester.
Alternative Flow	Alternative Flow - Panel still pending	
	3a.	Panel appointment is still pending approval.
	3b.	System displays the current status without showing unreleased details.
Exception Flow	Exception Flow - No panel record	
	3a.	System finds no panel appointment for the student.
	3b.	System displays a No panel record message.

UC23 - Generate Panel Confirmation Letter

Use Case Name	Generate Panel Confirmation Letter	
Use Case ID	UC23	
Description	The system auto-generates a panel appointment confirmation letter upon approval and delivers it to the student and all assigned panel members within the system.	
Priority	High	
Actor(s)	System, Student, Lecturer (Panel Member)	
Triggering Event	A panel appointment is approved through the workflow.	
Pre-condition	1. Panel appointment has been approved. 2. Student and panel member records exist. 3. Panel confirmation letter template is available.	
Post-condition	1. Panel confirmation letter is generated and saved to the document repository. 2. Letter is delivered to the student and all panel members within the system. 3. All recipients receive a notification.	
Flow of Events	1.	System detects that the panel appointment has been approved.
	2.	System retrieves the panel confirmation letter template.
	3.	System auto-fills the template with student and panel member details.
	4.	System generates the letter as a PDF file.
	5.	System saves the letter to the document repository.
	6.	System links the letter to the student and each panel member's records.
	7.	System delivers the letter notification to all recipients.
	8.	Recipients can preview or download the letter from their dashboard.
Alternative Flow	Alternative Flow - Manual regeneration after panel change	
	1a.	Office Staff regenerates the letter after a panel composition change.

	1b.	System regenerates the letter with updated panel members.
	1c.	System replaces the previous version and notifies all recipients.
Exception Flow	Exception Flow - Template missing or invalid	
	2a.	System cannot find the panel confirmation letter template.
	2b.	System logs the error and notifies Admin.
	2c.	System aborts the letter generation.

Module: Marks Entry and Management

UC24 - Configure Mark Entry Periods

Use Case Name	Configure Mark Entry Periods	
Use Case ID	UC24	
Description	The system allows authorized office staff to configure mark entry periods and deadlines per semester.	
Priority	High	
Actor(s)	Office Staff, System	
Triggering Event	Office Staff wants to set up or update mark entry periods for a semester.	
Pre-condition	1. Office Staff is logged into the system. 2. Office Staff has permission to configure mark entry periods.	
Post-condition	1. Mark entry period and deadlines are configured for the semester. 2. Configuration is recorded in the audit log.	
Flow of Events	1.	Office Staff opens the Mark Entry Configuration page.
	2.	Office Staff selects the target semester.
	3.	System displays the configuration form.
	4.	Office Staff enters the start date, end date, and deadlines.
	5.	Office Staff confirms the configuration.

	6.	System validates the date inputs.
	7.	System saves the mark entry period configuration.
	8.	System displays a success popup hint.
Alternative Flow	Alternative Flow - Reuse previous semester configuration	
	3a.	Office Staff selects the option to copy from a previous semester.
	3b.	System displays the previous configuration for editing.
	3c.	Office Staff adjusts the dates and saves the configuration.
Exception Flow	Exception Flow - Invalid date range	
	6a.	System detects that end date is before start date.
	6b.	System displays a validation error.
	6c.	Office Staff corrects the dates.

UC25 - Assign Evaluation Tasks

Use Case Name	Assign Evaluation Tasks
Use Case ID	UC25
Description	The system automatically assigns student evaluation tasks to approved panel members based on the configured mark entry period, approved panel appointment records, and defined rubric components.
Priority	High
Actor(s)	Office Staff, System
Triggering Event	Office Staff opens or activates the Mark Entry and Management section after mark entry periods and rubric components have been configured.
Pre-condition	1. Office Staff is logged into the system. 2. Mark entry period has been configured for the semester. 3. Mark entry period is currently open or activated. 4. Panel appointment records have been approved. 5. Mark rubric components are defined.
Post-condition	1. Evaluation tasks are assigned to specific panel members. 2. Assigned panel members receive a notification.

Flow of Events	1.	Office Staff opens the Mark Entry and Management section.
	2.	Office Staff selects the target semester or project stage.
	3.	System checks whether the mark entry period has been configured and is currently active.
	4.	System retrieves the approved panel appointment records for the selected semester or project stage.
	5.	System retrieves the defined mark rubric components.
	6.	System automatically generates evaluation tasks for each student based on their approved panel members.
	7.	System assigns the relevant evaluation tasks to each respective panel member.
	8.	System saves the generated evaluation task assignments.
	9.	System notifies the assigned panel members.
	10.	System records the automatic assignment activity in the audit log.
	11.	System displays the assignment summary to the Office Staff.
Alternative Flow	Alternative Flow - Existing assignments found	
	6a.	System detects that evaluation tasks have already been assigned for the selected semester or project stage.
	6b.	System displays the existing assignment summary.
	6c.	Office Staff may review the existing assignments without generating duplicate tasks.
Exception Flow	Exception Flow 1 - Incomplete assignment	
	3a.	System detects that the mark entry period or rubric components have not been configured.
	3b.	System displays a warning message indicating the missing configuration.
	3c.	Office Staff completes the required configuration before the system can assign evaluation tasks.
	Exception Flow 2 - No approved panel appointment found	
	4a.	System detects that a student does not have approved panel members.

	4b.	System excludes the student from automatic task assignment.
	4c.	System displays a warning listing students without approved panel appointments.

UC26 - Define Mark Rubric Components

Use Case Name	Define Mark Rubric Components	
Use Case ID	UC26	
Description	The system allows authorized office staff to define mark components or rubric items per semester or project stage.	
Priority	Medium	
Actor(s)	Office Staff, System	
Triggering Event	Office Staff wants to define or update mark rubric components.	
Pre-condition	1. Office Staff is logged into the system. 2. Office Staff has permission to manage mark rubric components.	
Post-condition	1. Mark rubric components are defined for the semester or project stage. 2. Rubric is available for use in automatic evaluation task assignment and mark entry.	
Flow of Events	1.	Office Staff opens the Mark Rubric Management page.
	2.	Office Staff selects the target semester or project stage.
	3.	System displays the rubric configuration form.
	4.	Office Staff enters rubric components with weight and maximum score.
	5.	Office Staff confirms the rubric.
	6.	System validates the rubric, including total weight and score ranges.
	7.	System saves the rubric.
	8.	System displays a success popup hint.
Alternative Flow	Alternative Flow - Reuse template	
	3a.	Office Staff selects a previously saved rubric template.

	3b.	System loads the template into the form for editing.
	3c.	Office Staff adjusts and saves the rubric.
Exception Flow	Exception Flow - Weights do not sum correctly	
	6a.	System detects that the sum of component weights is not 100%.
	6b.	System displays a validation error.
	6c.	Office Staff adjusts the weights.

UC27 -Save Draft Marks and Submit

Use Case Name	Save Draft Marks and Submit	
Use Case ID	UC27	
Description	The system allows panel members to enter marks for assigned students, save the marks as draft, and submit the marks when finalized.	
Priority	High	
Actor(s)	Lecturer (Panel Member), System	
Triggering Event	Panel member wants to enter marks for assigned students.	
Pre-condition	1. Panel member is logged into the system. 2. Evaluation tasks have been assigned. 3. Mark entry period is open. 4. Assigned rubric components are available.	
Post-condition	1. Draft marks are saved if the panel member chooses to save as draft. 2. Submitted marks are recorded if the panel member chooses to submit. 3. Submitted marks are no longer editable unless reopened by authorized staff.	
Flow of Events	1.	Panel member opens the Mark Entry page.
	2.	System displays the list of assigned students and rubric components.
	3.	Panel member selects a student.

	4.	System displays the rubric form for that student.
	5.	Panel member enters marks for each assigned component.
	6.	Panel member clicks Save Draft.
	7.	System validates the entered values.
	8.	System saves the marks as draft.
	9.	System displays a success popup hint.
	10.	Panel member reviews the draft marks.
	11.	Panel member clicks Submit Marks.
	12.	System validates that all required mark components are completed.
	13.	System displays a submission confirmation message.
	14.	Panel member confirms the submission.
	15.	System saves the marks as submitted.
	16.	System updates the mark entry status to Submitted.
	17.	System prevents further editing of the submitted marks.
	18.	System displays a submission success popup hint.
Alternative Flow	Alternative Flow 1 - Continue editing later	
	6a.	Panel member exits without finalizing.
	6b.	System retains the draft marks for later editing.
	6c.	Panel member can continue editing the draft marks while the mark entry period is still open.
	Alternative Flow 2 - Submit directly without saving draft first	
	6a.	Panel member clicks Submit Marks after entering the marks.
	6b.	System validates that all required mark components are completed.
	6c.	System displays a submission confirmation message.
	6d.	Panel member confirms the submission.
	6e.	System saves the marks as submitted and updates the status to Submitted.

Exception Flow	Exception Flow 1 - Mark entry period closed	
	7a.	System detects that the mark entry period has closed.
	7b.	System displays a Mark entry closed message.
	7c.	System prevents further mark entry.
	Exception Flow 2 - Incomplete marks during submission	
	12a.	System detects that required mark components are incomplete.
	12b.	System displays a validation error listing the incomplete components.
	12c.	Panel member completes the required marks before submission.
	Exception Flow 3 - Invalid mark value	
	7a.	System detects that the entered mark is outside the allowed score range.
	7b.	System displays a validation error.
	7c.	Panel member corrects the mark value.

UC28 - Validate Marks Before Submission

Use Case Name	Validate Marks Before Submission	
Use Case ID	UC28	
Description	The system validates mark completeness and acceptable score range before submission.	
Priority	High	
Actor(s)	Lecturer (Panel Member), System	
Triggering Event	Panel member attempts to submit marks for finalization.	
Pre-condition	1. Draft marks have been entered. 2. Mark rubric and ranges are defined.	
Post-condition	1. Marks are validated as complete and within range, or flagged for correction.	
Flow of Events	1.	Panel member clicks Submit on the mark entry page.
	2.	System retrieves the entered marks.

	3.	System checks completeness against the rubric.
	4.	System validates each mark is within the acceptable range.
	5.	System displays the validation result.
	6.	If valid, system enables the final submission.
Alternative Flow	None.	
Exception Flow	Exception Flow 1 - Missing marks	
	3a.	System detects missing marks for required components.
	3b.	System highlights the missing components.
	3c.	System prevents submission until marks are completed.
Exception Flow	Exception Flow 2 - Out-of-range marks	
	4a.	System detects marks outside the acceptable range.
	4b.	System highlights the invalid entries.
	4c.	Panel member corrects the values.

UC29 - View Mark Entry Records

Use Case Name	View Mark Entry Records	
Use Case ID	UC29	
Description	The system allows authorized users to view mark entries filtered by student, lecturer, semester, and status.	
Priority	Medium	
Actor(s)	Programme Coordinator, Office Staff, Lecturer (Panel Member), System	
Triggering Event	User wants to review mark entry records.	
Pre-condition	1. User is logged into the system. 2. Mark entry records exist. 3. User has permission to view marks.	
Post-condition	1. Matching mark entry records are displayed according to the user's role-based access permission.	
Flow of Events	1.	User opens the Mark Entry Records page.
	2.	System identifies the user's role and access permission.

	3.	System displays the filter interface.
	4.	User applies filters such as student, lecturer, semester, programme, or status.
	5.	System retrieves matching records based on the applied filters and the user's role-based access permission.
	6.	System displays the accessible mark entry records in a structured table.
	7.	User selects a record to view details.
	8.	System displays the mark entry details if the user is authorized to access the selected record.
Alternative Flow	Alternative Flow - Export records	
	6a.	User clicks Export.
	6b.	System exports only the mark entry records that the user is authorized to view.
	6c.	System provides the records as a downloadable file.
Exception Flow	Exception Flow 1 - No matching records	
	5a.	System finds no matching records.
	5b.	System displays a No records found message.
	Exception Flow 2 - Unauthorized record access	
	8a.	User attempts to access a mark entry record outside their permission scope.
	8b.	System denies access to the record.
	8c.	System displays an unauthorized access message.

Module: Workflow & Approval Tracking

UC30 - Route Supervisor Appointment Workflow

Use Case Name	Route Supervisor Appointment Workflow
Use Case ID	UC30
Description	The system routes supervisor appointment requests through the required approval workflow.

Priority	High	
Actor(s)	System, Lecturer (Supervisor), Programme Coordinator	
Triggering Event	Supervisor appointment request is submitted or processed.	
Pre-condition	1. Supervisor appointment request exists. 2. Approval workflow is configured.	
Post-condition	1. Request is routed to the next approver in the workflow. 2. Workflow status is updated.	
Flow of Events	1.	System receives the supervisor appointment request.
	2.	System identifies the current workflow stage.
	3.	System retrieves the next approver for the stage.
	4.	System updates the request status to the next stage.
	5.	System notifies the next approver of pending action.
	6.	Approver reviews the request and takes action.
	7.	System routes the request based on the approver's action.
Alternative Flow	Alternative Flow - Final approval reached	
	7a.	System detects that the final approval has been completed.
	7b.	System marks the request as Approved.
	7c.	System triggers downstream actions (e.g., letter generation).
Exception Flow	Exception Flow - Approver unavailable	
	5a.	System detects that the assigned approver is unavailable.
	5b.	System escalates or routes to a backup approver.
	5c.	System logs the escalation.

UC31 - Route Panel Appointment Workflow

Use Case Name	Route Panel Appointment Workflow
Use Case ID	UC31
Description	The system routes panel appointment recommendations through Programme Coordinator approval.

Priority	High	
Actor(s)	System, Lecturer (Panel member), Programme Coordinator	
Triggering Event	Panel recommendation is submitted by supervisor.	
Pre-condition	1. Panel recommendation exists. 2. Approval workflow is configured.	
Post-condition	1. Panel recommendation is routed to the Programme Coordinator. 2. Workflow status is updated.	
Flow of Events	1.	System receives the panel recommendation.
	2.	System identifies the current workflow stage.
	3.	System retrieves the next approver for the stage.
	4.	System updates the request status to the next stage.
	5.	System notifies the next approver of pending action.
	6.	Approver reviews the request and takes action.
	7.	System routes the request based on the approver's action.
Alternative Flow	Alternative Flow - Final approval reached	
	7a.	System detects that the final approval has been completed.
	7b.	System marks the request as Approved.
	7c.	System triggers downstream actions (e.g., letter generation).
Exception Flow	Exception Flow - Approver unavailable	
	5a.	System detects that the approver is unavailable.
	5b.	System escalates to backup approver.
	5c.	System logs the escalation.

UC32 - Manage Record Approval

Use Case Name	Manage Record Approval	
Use Case ID	UC32	
Description	The system allows approvers to approve or reject records at each workflow stage, with a mandatory comment required upon rejection.	
Priority	High	
Actor(s)	Approver (Lecturer, Programme Coordinator), System	
Triggering Event	Approver receives a record for review at their workflow stage.	
Pre-condition	1. Approver is logged into the system. 2. Record is at the approver's workflow stage.	
Post-condition	1. Record is approved or rejected. 2. Approval action with mandatory comment (on rejection) is recorded. 3. Workflow advances or terminates based on the decision.	
Flow of Events	1.	Approver opens the Pending Approvals page.
	2.	System displays the list of records awaiting their action.
	3.	Approver selects a record to review.
	4.	System displays the record details.
	5.	Approver reviews the information.
	6.	Approver clicks Approve or Reject.
	7.	If rejecting, system requires the approver to enter a mandatory comment.
	8.	Approver confirms the action.
	9.	System records the action in the audit log.
	10.	System updates the workflow status.
Alternative Flow	Alternative Flow - Add optional approval comment	
	6a.	Approver chooses to approve with an optional comment.
	6b.	System saves the comment with the approval.
Exception Flow	Exception Flow - Missing rejection comment	
	7a.	Approver attempts to reject without providing a comment.
	7b.	System displays a comment required error.

	7c.	System prevents the rejection until the comment is provided.
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UC33 - Maintain Workflow History

Use Case Name	Maintain Workflow History	
Use Case ID	UC33	
Description	The system maintains a workflow history for each workflow record, showing the stages, assigned approvers, actions taken, timestamps, and comments throughout the approval process.	
Priority	High	
Actor(s)	System, Lecturer(Supervisor/Panel), Programme Coordinator	
Triggering Event	Any workflow action is performed (submission, approval, rejection).	
Pre-condition	1. A workflow record exists. 2. A workflow action has occurred.	
Post-condition	1. Workflow history is updated with the latest action details. 2. Authorized users can view the workflow progress and history within the related record details.	
Flow of Events	1.	System detects a workflow action.
	2.	System captures the workflow stage, user, action type, timestamp, status, and comment.
	3.	System updates the workflow history for the related record.
	4.	User opens the related supervisor appointment or panel appointment record.
	5.	System displays the current workflow status.
	6.	System displays the workflow history as a stage-based timeline.
	7.	User reviews the completed, current, and pending workflow stages.
Alternative Flow	Alternative Flow - View detailed action record	
	6a.	User selects a workflow stage from the timeline.

	6b.	System displays detailed information for that stage, including approver, action, date, and comment.
Exception Flow	Exception Flow - Workflow history update failure	
	3a.	System fails to update the workflow history.
	3b.	System logs the failure to a fallback log.

Module: File Management

UC34 - Upload File to Repository

Use Case Name	Upload File to Repository	
Use Case ID	UC34	
Description	The system allows authorized users to upload academic and administrative files to the central repository, with metadata tags such as Student ID, Category, Module, and Semester.	
Priority	High	
Actor(s)	Office Staff, Programme Coordinator, Lecturer, System	
Triggering Event	User wants to upload a file into the central document repository.	
Pre-condition	1. User is logged into the system. 2. User has permission to upload files.	
Post-condition	1. File is stored in the central repository with metadata. 2. Upload action is recorded in the audit log.	
Flow of Events	1.	User opens the File Management module.
	2.	User clicks the Upload File button.
	3.	System displays the file upload form with file selector and metadata fields.
	4.	User selects the file from their local device.
	5.	User enters metadata tags (Student ID, Category, Module, Semester).
	6.	User confirms the upload.
	7.	System validates the file size, format, and metadata.
	8.	System uploads the file to the repository storage.

	9.	System creates a file metadata record.
	10.	System displays a success popup hint with the file reference ID.
Alternative Flow	Alternative Flow - Drag and drop upload	
	4a.	User drags the file into the upload zone.
	4b.	System auto-fills the file name field.
	4c.	User completes any remaining metadata fields and confirms upload.
Exception Flow	Exception Flow 1 - File too large	
	7a.	System detects the file exceeds the maximum allowed size.
	7b.	System displays a File size exceeds maximum allowed error.
	7c.	User must compress the file or upload a smaller file.
Exception Flow	Exception Flow 2 - Unsupported file type	
	7a.	System detects an unsupported file extension.
	7b.	System displays an Unsupported file type error message.
	7c.	User must convert the file or upload a supported format.

UC35 - Upload Student Documents

Use Case Name	Upload Student Documents
Use Case ID	UC35
Description	The system allows students to upload required academic or administrative documents to the repository under their own student record.
Priority	High
Actor(s)	Student, System
Triggering Event	Student wants to submit a document required by the postgraduate office.
Pre-condition	1. Student is logged into the system. 2. Student has permission to upload documents. 3. Accepted file types and size limits are configured.

Post-condition	1. Document is uploaded and linked to the student's record. 2. Upload action is recorded in the audit log.	
Flow of Events	1.	Student opens the Document Upload page.
	2.	System displays the upload interface and required metadata fields.
	3.	Student selects a file.
	4.	Student enters category, module, semester, and notes if required.
	5.	System validates file type, file size, and metadata.
	6.	Student confirms upload.
	7.	System stores the file and links it to the student profile.
	8.	System displays a success popup hint.
Alternative Flow	Alternative Flow - Drag and drop upload	
	3a.	Student drags and drops the file into the upload zone.
	3b.	System auto-fills the file name and continues with metadata entry.
Exception Flow	Exception Flow 1 - Unsupported file or oversized file	
	5a.	System detects unsupported file type or excessive file size.
	5b.	System displays the issue.
	5c.	System prevents upload until corrected.

UC36 - Search and Filter Files

Use Case Name	Search and Filter Files
Use Case ID	UC36
Description	The system allows authorized users to search and filter files by filename, metadata, uploader, date, or keyword.
Priority	Medium
Actor(s)	Admin, Office Staff, Programme Coordinator, Lecturer, System
Triggering Event	User wants to find specific files in the repository using search or filter criteria.

Pre-condition	1. User is logged into the system. 2. User has permission to view files.	
Post-condition	1. List of matching files is displayed. 2. User can select a file for further action.	
Flow of Events	1.	User opens the File Management module.
	2.	System displays the search and filter interface.
	3.	User enters a search keyword or selects filter criteria.
	4.	User submits the search or filter request.
	5.	System queries the repository with the specified criteria.
	6.	System retrieves matching files based on permissions.
	7.	System displays the filtered file list.
	8.	User reviews the results or selects a file for further action.
Alternative Flow	Alternative Flow - Combine search with filter	
	3a.	User enters a keyword and selects multiple filter criteria.
	3b.	System combines the search and filter criteria.
	3c.	System retrieves files matching all the specified criteria.
Exception Flow	Exception Flow 1 - No matching files	
	6a.	System finds no files matching the criteria.
	6b.	System displays a No matching files found message.

UC37 - Preview and Download File

Use Case Name	Preview and Download File
Use Case ID	UC37
Description	The system allows authorized users to preview supported file types in the browser and download authorized files and generated letters from the repository.
Priority	High
Actor(s)	Admin, Office Staff, Programme Coordinator, Lecturer, Student, System
Triggering Event	User wants to view or download a file from the repository.

Pre-condition	1. User is logged into the system. 2. User has permission to access the target file. 3. Target file exists in the repository.	
Post-condition	1. File is previewed in the browser or downloaded to the user's device. 2. Access action is recorded in the audit log.	
Flow of Events	1.	User opens the File Management module.
	2.	User locates the desired file using search or filter.
	3.	User clicks the file name to preview or clicks the download icon.
	4.	System verifies the user's access permissions for the file.
	5.	System retrieves the file from storage.
	6.	System renders the file preview or initiates the download.
	7.	User views or saves the file.
Alternative Flow	Alternative Flow - Bulk download	
	3a.	User selects multiple files using checkboxes.
	3b.	User clicks Bulk Download.
	3c.	System packages the files into a compressed archive and initiates download.
Exception Flow	Exception Flow 1 - Access denied	
	4a.	System detects that the user lacks permission.
	4b.	System displays a Permission denied message.
	4c.	System prevents the preview or download.
Exception Flow	Exception Flow 2 - File missing	
	5a.	System cannot locate the file in storage.
	5b.	System displays a File not found error.
	5c.	System notifies Admin.

UC38 - Manage File Access Control

Use Case Name	Manage File Access Control
Use Case ID	UC38

Description	The system allows the Admin to enforce role-based file access control by module and action, and to link files to related student records and workflow records.	
Priority	High	
Actor(s)	Admin, System	
Triggering Event	Admin wants to configure file access permissions or link files to records.	
Pre-condition	1. Admin is logged into the system. 2. Admin has permission to manage file access controls. 3. Target files or file categories exist.	
Post-condition	1. File access permissions are updated. 2. Files are linked to related records. 3. Changes are recorded in the audit log.	
Flow of Events	1.	Admin opens the File Access Control page.
	2.	System displays the access control configuration.
	3.	Admin selects a file or file category.
	4.	System displays the current role-based permission matrix.
	5.	Admin modifies the permissions for each role and action.
	6.	Admin links the file to related student or workflow records.
	7.	Admin confirms the changes.
	8.	System validates the configuration.
	9.	System applies the new access control rules.
	10.	System displays a success popup hint.
Alternative Flow	Alternative Flow - Apply default permission template	
	5a.	Admin selects a default permission template.
	5b.	System applies the template to the selected file or category.
Exception Flow	Exception Flow 1 - Conflicting rules	
	8a.	System detects conflicting permission rules.
	8b.	System displays a warning.
	8c.	Admin reviews and resolves the conflicts.

UC39 - Archive Old Files

Use Case Name	Archive Old Files	
Use Case ID	UC39	
Description	The system allows authorized users to archive old or inactive files to maintain repository cleanliness.	
Priority	Medium	
Actor(s)	Office Staff, Admin, System	
Triggering Event	User wants to archive an old or inactive file.	
Pre-condition	1. User is logged into the system. 2. User has permission to archive files. 3. Target file exists in the active repository.	
Post-condition	1. File is moved to the archive section. 2. File no longer appears in the active file list. 3. Archive action is recorded in the audit log.	
Flow of Events	1.	User opens the File Management module.
	2.	User locates and selects the file to archive.
	3.	User clicks the Archive option.
	4.	System displays a confirmation dialog.
	5.	User confirms the archive action.
	6.	System updates the file status to archived.
	7.	System moves the file metadata to the archive section.
	8.	System displays a success popup hint.
Alternative Flow	Alternative Flow - Restore archived file	
	1a.	User opens the archive section.
	1b.	User selects an archived file and clicks Restore.
	1c.	System updates the file status back to active.
Exception Flow	Exception Flow 1 - File linked to active workflow	
	6a.	System detects the file is linked to an active workflow.
	6b.	System displays a warning.
	6c.	User reviews and decides whether to proceed.

Module: Letter Generation

UC40 - Manage Letter Templates

Use Case Name	Manage Letter Templates	
Use Case ID	UC40	
Description	The system allows the Admin to maintain reusable letter templates for various letter types (Visa Support, Confirmation, Supervisor Confirmation, Panel Confirmation, etc.).	
Priority	High	
Actor(s)	Admin, System	
Triggering Event	Admin wants to create, edit, or archive a letter template.	
Pre-condition	1. Admin is logged into the system. 2. Admin has permission to manage letter templates.	
Post-condition	1. Letter template list is updated. 2. Active templates are available for letter generation. 3. Template change is recorded in the audit log.	
Flow of Events	1.	Admin opens the Letter Generation module.
	2.	Admin selects Manage Letter Templates.
	3.	System displays the list of existing templates with statuses.
	4.	Admin selects an action (create, edit, or archive).
	5.	Admin uploads or edits the template content with placeholders.
	6.	Admin defines the placeholder-to-field mapping.
	7.	Admin confirms the changes.
	8.	System validates the template structure and placeholders.
	9.	System updates the template record.
	10.	System displays a success popup hint.
Alternative Flow	Alternative Flow - Preview template	
	5a.	Admin selects Preview after editing.

Exception Flow	5b.	System renders the template with sample student data.
	5c.	Admin reviews and adjusts if needed.
	Exception Flow 1 - Invalid placeholders	
	8a.	System detects unknown placeholders.
	8b.	System displays the invalid placeholders.
	8c.	Admin corrects and resubmits.

UC41 - Generate Standardized Letter

Use Case Name	Generate Standardized Letter	
Use Case ID	UC41	
Description	The system allows office staff to generate standardized letters with auto-fill from the Student Registry, manual editing of selected fields, and export to PDF or Word format.	
Priority	High	
Actor(s)	Office Staff, System	
Triggering Event	Office Staff wants to generate an official letter for a student.	
Pre-condition	1. Office Staff is logged into the system. 2. Office Staff has permission to generate letters. 3. Letter templates are available. 4. Target student record exists.	
Post-condition	1. Letter is generated in PDF or Word format. 2. Reference number is assigned to the letter. 3. Letter is saved to the document repository. 4. Letter generation is recorded in the audit log.	
Flow of Events	1.	Office Staff opens the Letter Generation module.
	2.	System displays available letter types.
	3.	Office Staff selects a letter type.
	4.	Office Staff searches and selects the target student.
	5.	System retrieves student data and auto-fills the template.
	6.	System displays the auto-filled letter preview.
	7.	Office Staff reviews the preview.

	8.	Office Staff optionally edits specific editable fields.
	9.	Office Staff confirms generation.
	10.	System assigns a unique reference number.
	11.	System renders the letter in the selected format (PDF or Word).
	12.	System saves the letter to the document repository.
	13.	System provides a download link.
Alternative Flow	Alternative Flow - Manual edit of letter fields	
	7a.	Office Staff selects fields to edit manually.
	7b.	System enables inline editing for those fields.
	7c.	Office Staff modifies the values and saves.
Exception Flow	Exception Flow 1 - Missing student data	
	5a.	System detects required student fields are missing.
	5b.	System displays a warning listing the missing fields.
	5c.	Office Staff fills in manually or updates the Student Registry.
Exception Flow	Exception Flow 2 - Generation failure	
	11a.	System encounters an error during letter rendering.
	11b.	System logs the error and displays a Generation failed message.
	11c.	Office Staff retries or contacts technical support.

UC42 - Deliver Letter to Recipients

Use Case Name	Deliver Letter to Recipients
Use Case ID	UC42
Description	The system delivers generated letters in-system to relevant recipients (student, supervisor, panel members) with notifications.
Priority	High
Actor(s)	System, Student, Lecturer (Supervisor, Panel Member)
Triggering Event	A letter is generated and ready for delivery.

Pre-condition	1. Letter has been generated and saved to the repository. 2. Recipient list is identified.	
Post-condition	1. Letter is linked to each recipient's record. 2. Each recipient receives a notification. 3. Recipients can preview or download the letter from their dashboard.	
Flow of Events	1.	System receives the generated letter for delivery.
	2.	System identifies the recipient list (based on letter type and context).
	3.	System links the letter to each recipient's record.
	4.	System creates an in-app notification for each recipient.
	5.	System updates the recipient's letter inbox.
	6.	Recipients open the notification and access the letter.
Alternative Flow	Alternative Flow - Resend letter notification	
	5a.	Office Staff manually triggers a resend.
	5b.	System creates a new notification for the affected recipients.
Exception Flow	Exception Flow 1 - Recipient record missing	
	3a.	System cannot find one or more recipient records.
	3b.	System logs the failure and notifies Office Staff.
	3c.	Office Staff resolves the missing record and retries.

Module: Chatbot

UC43 - Submit FAQ Query

Use Case Name	Submit FAQ Query
Use Case ID	UC43
Description	The system accepts and answers natural language FAQ queries from logged-in students using the published knowledge base.
Priority	High
Actor(s)	Student, System

Triggering Event	Student wants to ask a question about postgraduate office procedures.	
Pre-condition	1. Student is logged into the system. 2. Chatbot service is running. 3. FAQ knowledge base contains published entries.	
Post-condition	1. Student receives an answer or is advised to refer to the postgraduate counter. 2. Query is logged for analysis.	
Flow of Events	1.	Student opens the chatbot interface.
	2.	System displays the chatbot welcome message.
	3.	Student types a question in natural language.
	4.	System validates the input.
	5.	System forwards the query to the FAQ search engine.
	6.	System searches the published knowledge base.
	7.	System retrieves the best matching FAQ entry.
	8.	System displays the answer to the student.
	9.	Student reviews the answer or submits another query.
Alternative Flow	Alternative Flow - Follow-up question	
	8a.	Student asks a follow-up question.
	8b.	System processes and returns the next answer.
Exception Flow	Exception Flow 1 - No reliable answer	
	6a.	System finds no matching FAQ entry above the relevance threshold.
	6b.	System advises the student to refer to the postgraduate counter.
	6c.	System logs the unanswered query for review.

UC44 - Manage FAQ Knowledge Base

Use Case Name	Manage FAQ Knowledge Base
Use Case ID	UC44

Description	The system allows the Admin to manage the FAQ knowledge base by creating, editing, uploading reference documents, archiving, and deleting FAQ records.	
Priority	Medium	
Actor(s)	Admin, System	
Triggering Event	Admin wants to update or clean up the FAQ knowledge base.	
Pre-condition	1. Admin is logged into the system. 2. Admin has permission to manage the FAQ knowledge base.	
Post-condition	1. FAQ knowledge base is updated. 2. Only published entries are searchable by the chatbot. 3. Changes are recorded in the audit log.	
Flow of Events	1.	Admin opens the FAQ Knowledge Base page.
	2.	System displays the list of FAQ entries with statuses.
	3.	Admin selects an action (create, edit, publish, archive, or delete).
	4.	Admin enters or modifies the question, answer, and tags.
	5.	Admin confirms the changes.
	6.	System validates the content.
	7.	System updates the FAQ record.
	8.	System displays a success popup hint.
Alternative Flow	Alternative Flow - Upload reference documents	
	3a.	Admin selects Upload Reference Documents.
	3b.	Admin uploads PDF or DOCX files.
	3c.	System validates and indexes the documents.
Exception Flow	Exception Flow 1 - Empty question or answer	
	6a.	System detects an empty question or answer field.
	6b.	System displays a Required field error.
	6c.	Admin completes the fields.

Module: Announcement

UC45 - Publish Announcement

Use Case Name	Publish Announcement	
Use Case ID	UC45	
Description	The system allows authorized users to publish announcements with target audience selection, schedule (start/expiry), and attachments.	
Priority	High	
Actor(s)	Admin, Office Staff, System	
Triggering Event	User wants to broadcast an announcement.	
Pre-condition	1. User is logged into the system. 2. User has permission to publish announcements.	
Post-condition	1. Announcement record is created and published or scheduled. 2. Notifications are triggered for the targeted users. 3. Action is recorded in the audit log.	
Flow of Events	1.	User opens the Announcement module.
	2.	User clicks Create Announcement.
	3.	System displays the composition form.
	4.	User enters title, content, target audience, schedule, and attachments.
	5.	User clicks Publish Now or Schedule.
	6.	System validates the content.
	7.	System creates the announcement record.
	8.	If Publish Now, system triggers notifications for targeted users.
	9.	If Scheduled, system queues the announcement for the start time.
	10.	System displays a success popup hint.
Alternative Flow	Alternative Flow - Save as draft	
	5a.	User clicks Save as Draft.
	5b.	System saves with draft status.
	5c.	No notifications are triggered.

Exception Flow	Exception Flow 1 - Missing required fields	
	6a.	System detects missing title or content.
	6b.	System highlights the missing fields.
	6c.	User completes them.
Exception Flow	Exception Flow 2 - Invalid schedule dates	
	6a.	System detects that publish date is after expiry date.
	6b.	System displays a validation error.
	6c.	User corrects the dates.

Module: Dashboard & Semester Timeline

UC46 - View Semester Timeline and Tasks

Use Case Name	View Semester Timeline and Tasks	
Use Case ID	UC46	
Description	The system displays the active semester timeline, key academic deadlines, and pending task lists specific to each user's role.	
Priority	High	
Actor(s)	All authenticated users, System	
Triggering Event	User opens the dashboard.	
Pre-condition	1. User is logged into the system. 2. Semester timeline data is uploaded. 3. User role is identified.	
Post-condition	1. Active semester timeline is displayed. 2. User's role-specific pending tasks are shown.	
Flow of Events	1.	User opens the dashboard.
	2.	System identifies the user's role.
	3.	System retrieves the active semester timeline.
	4.	System retrieves pending tasks specific to the user's role.
	5.	System displays the timeline and task list on the dashboard.

	6.	User selects a task to navigate to the related record.
Alternative Flow	Alternative Flow - No pending tasks	
	4a.	System finds no pending tasks for the user.
	4b.	System displays the timeline with an empty task list message.
Exception Flow	Exception Flow - Timeline not available	
	3a.	System cannot retrieve the semester timeline.
	3b.	System displays "No timeline available at now" and keeps other dashboard sections available.

UC47 - Upload Semester Timeline

Use Case Name	Upload Semester Timeline	
Use Case ID	UC47	
Description	The system allows Admin to upload semester timeline data from a structured Excel template, with validation of the template structure.	
Priority	High	
Actor(s)	Admin, System	
Triggering Event	Admin wants to upload the timeline for a new semester.	
Pre-condition	1. Admin is logged into the system. 2. Admin has the prepared Excel template filled with timeline data.	
Post-condition	1. Semester timeline data is imported into the system. 2. Timeline becomes available on dashboards. 3. Upload action is recorded in the audit log.	
Flow of Events	1.	Admin opens the Semester Timeline module.
	2.	Admin clicks Upload Timeline.
	3.	System displays the upload interface with a template download link.
	4.	Admin uploads the filled Excel file.
	5.	System validates the file structure and required fields.
	6.	System parses the timeline rows.
	7.	System imports the timeline data.

	8.	System displays an import summary.
	9.	System displays a success popup hint.
Alternative Flow	Alternative Flow - Download template	
	3a.	Admin clicks the template download link.
	3b.	System provides the Excel template for the timeline data.
Exception Flow	Exception Flow - Invalid template structure	
	5a.	System detects missing required columns or invalid structure.
	5b.	System displays the validation errors.
	5c.	Admin corrects the file and re-uploads.

UC48 - Manage Semester Timeline

Use Case Name	Manage Semester Timeline	
Use Case ID	UC48	
Description	The system allows Admin to update or replace the active semester timeline as the semester progresses.	
Priority	Medium	
Actor(s)	Admin, System	
Triggering Event	Admin wants to update or replace the active semester timeline.	
Pre-condition	1. Admin is logged into the system. 2. An active semester timeline exists.	
Post-condition	1. Active semester timeline is updated or replaced. 2. Change is reflected on all dashboards. 3. Change is recorded in the audit log.	
Flow of Events	1.	Admin opens the Semester Timeline module.
	2.	System displays the current active timeline.
	3.	Admin selects Update or Replace.
	4.	Admin uploads a new Excel file or edits the existing timeline entries.
	5.	Admin confirms the changes.

	6.	System validates the new data.
	7.	System updates or replaces the active timeline.
	8.	System displays a success popup hint.
Alternative Flow	Alternative Flow - Edit individual timeline entry	
	3a.	Admin selects an individual timeline entry to edit.
	3b.	Admin modifies the entry details.
	3c.	System validates and saves the change.
Exception Flow	Exception Flow - Conflicting timeline data	
	6a.	System detects conflicts between the new and existing data.
	6b.	System displays the conflicts.
	6c.	Admin resolves the conflicts before proceeding.

Module: Notification & Feedback

UC49 - Manage In-App Notifications

Use Case Name	Manage In-App Notifications	
Use Case ID	UC49	
Description	The system displays in-app notifications via the bell icon for system events such as approvals, deadlines, and announcements.	
Priority	High	
Actor(s)	All authenticated users, System	
Triggering Event	User clicks the notification bell icon or a system event triggers a notification.	
Pre-condition	1. User is logged into the system. 2. Notification service is running.	
Post-condition	1. User views the notification list. 2. Notifications are marked as read when opened.	
Flow of Events	1.	System detects a triggering event.

	2.	System creates a notification record for the targeted users.
	3.	System displays a notification indicator on the bell icon.
	4.	User clicks the bell icon.
	5.	System retrieves the user's recent notifications.
	6.	System displays the notification dropdown.
	7.	User selects a notification.
	8.	System opens the related record and marks the notification as read.
Alternative Flow	Alternative Flow - Mark all as read	
	6a.	User clicks Mark All as Read.
	6b.	System updates all unread notifications to read.
Exception Flow	Exception Flow 1 - No notifications	
	5a.	System finds no notifications for the user.
	5b.	System displays a No new notifications message.

UC50 - Display System Hint Popups

Use Case Name	Display System Hint Popups	
Use Case ID	UC50	
Description	The system displays real-time success or error popup hints for every system action to confirm execution result.	
Priority	High	
Actor(s)	All authenticated users, System	
Triggering Event	User performs any system action that returns a success or error response.	
Pre-condition	1. User is logged into the system. 2. User performs an action that requires feedback.	
Post-condition	1. Hint popup is displayed for a short duration. 2. Popup auto-dismisses or can be manually dismissed.	
Flow of Events	1.	User performs a system action.
	2.	System processes the action.

	3.	System receives the action result.
	4.	System displays a hint popup with appropriate styling (green for success, red for error).
	5.	Popup shows a clear message describing the result.
	6.	Popup auto-dismisses after the configured duration.
Alternative Flow	Alternative Flow - Manual dismiss	
	5a.	User clicks the close icon on the popup.
	5b.	System dismisses the popup immediately.
Exception Flow	Exception Flow 1 - Action failure	
	3a.	System detects an error during processing.
	3b.	System displays an error popup with a descriptive message.
	3c.	User reviews and retries or contacts support.